Flying inside the exhaust band

Ideal-then-throttle versus banded-optimal transfers for the SpaceTrains plasma fleet

SpaceTrains trajectory study

October 2026

Abstract

SpaceTrains plans its plasma ships with an *ideal* constant-power drive whose exhaust speed may take any value. The game's drives are confined to an exhaust band of 50–250 km/s. Two ways of respecting the band are compared on the game's five plasma hulls (four tank sizes each), eight heliocentric routes and 18–36 departure angles per route. **A** plans the ideal path and flies it inside the band, throttling at 250 km/s where the path asks for less thrust than the drive gives at full power. **B** plans the true optimum inside the band, with a new solver written for this study. At the same transfer time A burns a median 1.0% (mean 3.2%, maximum 45%) more propellant than B. The cargo hulls lose about 1%. The liners lose 6–7%, because their power-to-mass ratio puts their thrust at 250 km/s above what most of the path asks for. On a full tank B's shortest trip is 3.4% shorter than A's on average, and up to 40% shorter on short inner-system windows (there A takes up to 67% longer). The cheapest trip costs 1.95% more with A on average, up to 31.2% for the liners. B's paths coast for 15% (cargo) to 35–37% (liners) of the trip and leave A's path by 10^6–10^8 km on the outer legs. The cheapest tank differs between A and B on 12% of the departures (22% for the liners, 5% for the cargo hulls). Every B plan is checked by re-propagation at 10^{-12} and rejected if it does not reproduce. B solves in about 1 ms when it converges. It still fails on 16.1% of the fixed-time problems in the sweeps, and a failing search can take a minute. A cheaper exact-only B, used only to refine the trip A picks, recovers 1.04% of trip cost on average. The provisional recommendation is to keep A as the game's planner, with that refinement as an option.

1 The question

A constant-power drive with jet power P trades thrust for exhaust speed: at full power the thrust acceleration is $a = 2P/(mv_e)$. With v_e free, minimum propellant is minimum $\int a^2\,dt$, the optimal path does not depend on the ship's mass, and one atlas of paths serves every ship. This is how the game plans today.

The game's drives have an exhaust band $[v_{\min}, v_{\max}] = [50, 250]$ km/s. Writing $s = 2P/m$, at full power the band allows $a \in [a_{\min}, a_{\max}] = [s/v_{\max}, s/v_{\min}]$. Below $a_{\min}$ the drive can only run at $v_{\max}$ and part power (or pulse at full power, the same on average). Above $a_{\max}$ it cannot go at all.

A: plan ideal, fly banded. Plan the ideal path. Fly it exactly, throttled at $v_{\max}$ where $a < a_{\min}$. Reject it where $a > a_{\max}$. The path and its boundary conditions are unchanged; only the propellant grows. This is what the branch `feature/exhaust_band` does in the game.

B: plan banded, fly banded. Solve the minimum-propellant problem with the band as a constraint, and fly that.

Both are exact in the sense that the ship arrives where and when the plan says. The true banded optimum is never worse than A: it is the optimum among all controls the drive can fly, and A's control is one of them. A solver, though, finds only a local candidate, which can miss or land elsewhere; the study keeps that distinction (Section 4). In the same way the ideal engine is a lower bound for both, since $a\max(a, a_{\min}) \geq a^2$, but only its global optimum is; the ideal solver's answer is a candidate too. The questions are how much A loses (propellant at a fixed time, time on a fixed tank, cost of the trip), how different the paths are, whether the choice of tank changes, and whether B is fast enough for the game.

2 Model

Planar two-body motion about the Sun, normalised units ($\mu = 1$, length 1 AU, speed 29.78 km/s, time 58.13 days). The state is $(\mathbf{r}, \mathbf{v}, s)$ with

$$\dot{\mathbf{r}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = -\mathbf{r}/r^3 + a\,\hat{\mathbf{u}}, \qquad \dot{s} = l(a) = a\,\max(a, a_{\min}), \qquad a_{\min} = s/v_{\max}, \quad 0 \leq a \leq a_{\max} = s/v_{\min}. \tag{1}$$

Inside the band $l = a^2$ (the ideal drive). Throttled at $v_{\max}$ it is $l = a\,a_{\min}$, which is $\dot m = -ma/v_{\max}$, the rocket equation at $v_{\max}$. The final mass is $m_f = 2P/s(T)$, so minimum propellant is minimum $s(T)$. A ship enters only through $s_0 = 2P/m_0$ and its band. The ideal solution is the case $v_{\min} = 0$, $v_{\max} = \infty$, where $s(T) - s_0 = \int a^2 dt$ is the effort the game's atlas stores.

2.1 Optimal control inside the band

With the primer $\mathbf{p} = -\boldsymbol{\lambda}_v/2$ (so that the ideal thrust is $\mathbf{a} = \mathbf{p}$, the game solver's convention), $\mathbf{w} = \dot{\mathbf{p}}$, $\dot{\mathbf{w}} = G(\mathbf{r})\,\mathbf{p}$, and the costate λ_s of s, Pontryagin's principle asks the thrust to point along $\mathbf{p}$ and its size to minimise $-2|\mathbf{p}|\,a + \lambda_s\,l(a)$ on $[0, a_{\max}]$. With $q = |\mathbf{p}|/\lambda_s$:

arc	condition	thrust	$\dot\lambda_s$
coast	$q \le a_{\min}/2$	0	0
ceiling	$a_{\min}/2 < q < a_{\min}$	$a_{\min}$ (full power, $v_{\max}$)	$(2\lvert\mathbf{p}\rvert - 2\lambda_s a_{\min})/v_{\max}$
inside the band	$a_{\min} \le q < a_{\max}$	q	0
floor	$q \ge a_{\max}$	$a_{\max}$ (full power, $v_{\min}$)	$(2\lvert\mathbf{p}\rvert - 2\lambda_s a_{\max})/v_{\min}$

Below the ceiling the cost is linear in a, so the optimum never throttles: it burns at full power and $v_{\max}$ or coasts, and the switch between them is a jump in thrust. The costates are homogeneous, so $\lambda_s(0) = 1$ fixes their scale. The unknowns are then $(\mathbf{p}_0, \mathbf{w}_0)$, as for the ideal problem. With no band, $\lambda_s \equiv 1$ and the ideal primer solves the banded problem as it is. The Hamiltonian $H = 2\mathbf{w}\cdot\mathbf{v} - 2\mathbf{p}\cdot(\mathbf{g}+\mathbf{a}) + \lambda_s l(a)$ is constant along an extremal, which the tests use as a check.

3 The solver

`src/variable_isp/BandedTransfer.{hpp,cpp}` (namespace `variable_isp::banded`) is separate from the game's ideal solver. It uses only that solver's public types and its `solve_transfer` for the ideal seed. It is built as its own library, which only its test and the study driver link.

Propagation. Adaptive DOP853 (or DOPRI5), on one arc at a time. When the arc at the end of an accepted step differs from the arc at its start, the switch is located on the boundary function (Illinois iterations on trial steps in the old arc's flow). The step is cut there and the new arc continues. An arc can also be left and re-entered within one step (a grazing excursion), with the same arc at both ends. To catch it, the arc is also checked at seven interior points of the step's cubic Hermite interpolant; if one is on another arc, the step is retried ending there, so the crossing is located as above. An excursion shorter than an eighth of a step can still be missed; the re-propagation check (Section 5) catches the plans where that matters. The sensitivities of the end state to $(\mathbf{p}_0, \mathbf{w}_0)$ come from forward-mode dual numbers, four directions in one propagation. At a switch where the thrust jumps (coast/ceiling), the saltation term $\delta\mathbf{x}^+ = \delta\mathbf{x}^- + (\mathbf{f}^- - \mathbf{f}^+)\,\delta t^*$ carries the moving switch time into them.

Fixed-time solve. Damped Newton on the four rendezvous residuals. It works in up to three stages:

1. From a warm start (a neighbouring time or departure), Newton directly at the full band.
2. Otherwise continuation in band tightness: the band at $v_{\max}/\lambda$ and $\lambda v_{\min}$, λ from 0 (the ideal solution) to 1, adaptive steps and a secant predictor.
3. If that fails (typically when a coast arc is born on the way: there the end state depends on the unknowns through a square root and Newton stalls), the same continuation on a smoothed problem. The ceiling cost $a\max(a, a_{\min})$ is replaced by $a \cdot \frac{1}{2}\big(a + a_{\min} + \sqrt{(a-a_{\min})^2 + \varepsilon^2 a_{\min}^2}\big)$, whose exact minimiser is a continuous control. Then ε goes from 0.3 down to 10^{-2}, and an exact polish ($\varepsilon = 0$) follows. When a switch grazes its boundary the exact problem is ill-conditioned; the smoothed plan is then kept. Its intermediate thrust is throttling at $v_{\max}$, so it is a flyable, near-optimal plan, and its propellant is integrated with the drive's true flow.

A converged extremal that burns more than A at the same time is on another branch and is rejected.

Tuning. Variants were run on a tuning set: the five hulls, the eight routes, six departures each, and transfer times of 0.25, 0.5 and 1 Hohmann times, 720 cases in all (Table 1). Where several variants converge they land on the same extremal. What differs is how often they converge and at what cost. "A flies" marks the 399 cases where the ideal path stays under the 50 km/s floor, so a banded solution certainly exists. In the other 321, A cannot fly at all and B may or may not exist.

- **Speed.** A solve that converges usually takes 1–3 ms (the median). The mean is pulled up by a tail of hard cases.
- **Jacobian.** Dual numbers beat finite differences on both counts: 20 more cases, and half the propagations. A finite-difference step across a switch sees the switch move and returns a poor Jacobian.
- **Integrator.** DOP853 at 10^{-10} is kept. DOPRI5 solves three more cases but is 50% slower per solve. A looser 10^{-9} loses six cases to noisy switch times.
- **Continuation.** The secant predictor matters most: without it 50 cases are lost. Going to the full band in one step is as good on this set, because most cases need no continuation at all. The adaptive path is kept for the cases that do.
- **Smoothing fallback.** It adds 7 cases (2%), but it is expensive: a case that fails anyway costs about 1.2 s before giving up, against 50 ms without the fallback. The study keeps it for coverage. A game would not; it would fail fast and fly A instead (Section 6).

The cases still unsolved where A flies (49 of 399) are mostly long outer transfers near the Hohmann time. Their optimum is dominated by coasting, and coast arcs are born and die along the continuation path. On the other side, B solves 107 cases in which the ideal path breaks the floor and A cannot fly.

Table 1: Solver variants on the tuning set (cases where the ideal problem solves). One process per variant, eight in parallel, so absolute times are indicative; Table 7 has the one-thread timings.

Variant	solved	A flies: solved	median ms, solved	mean ms, solved	mean ms, failed	propagations mean
default	457/720	350/399	1.7	50	1169	228
no smoothing fallback	449/720	343/399	1.6	15	52	134
finite-difference Jacobian	435/720	330/399	2.7	29	289	414
DOPRI5 (1e-10)	459/720	353/399	2.5	67	611	221
DOP853 at 1e-9	451/720	344/399	1.6	60	1208	239
fallback smoothing down to 1e-3	456/720	349/399	2.0	54	1205	229
band in one step	458/720	352/399	1.1	43	1230	233
no secant predictor	407/720	300/399	2.0	95	1151	191

Validation. `tests/banded_transfer_test.cpp`:

- **No band, or a band the path never touches:** the ideal primer meets the target and $s(T) - s_0$ equals the ideal effort (to 10^{-8}).
- **Jacobian:** the dual-number Jacobian across switches matches finite differences.
- **Free-space rest-to-rest with a ceiling:** the solver finds burn, coast, burn, with switch times (10^{-7}) and final mass (10^{-9}) equal to the closed form. The closed form is $u = 1 - e^{-\Delta v/v_{\max}}$, $t_1 = u v_{\max}^2/s_0$, $t_2 = (1-u)t_1$, $s(T) = s_0/(1-u)^2$.
- **Throttled ideal path:** it matches the rocket equation at $v_{\max}$.
- **Hamiltonian:** constant across switches.
- **Heliocentric leg:** B never burns more than A, and a warm start from the answer converges at once.

4 Study setup

Ships. The five plasma hulls of the game (Table 2), each with its four tank variants (50, 100, 150 and 200% of the standard tank). Freighters and tankers carry a full hold at 100 kg a unit. Liners carry full berths, at 100 kg a passenger with luggage plus 100 days of consumables. The band is 50–250 km/s for all; police ships are out of scope. The Luna ferry only flies planet-system spirals and is left out.

Table 2: The plasma hulls at their standard tanks, fully loaded. $a_{\min}$ and $a_{\max}$ are the thrust accelerations at full power and 250 or 50 km/s, at departure. The daily cost is the game's capital charge plus crew and life support.

Ship	dry (t)	tank (t)	payload (t)	m_0 (t)	power (MW)	$a_{\min}$ (mm/s^2)	$a_{\max}$ (mm/s^2)	cr/day
Freighter	34.1	30	15.0	79.1	10.0	1.01	5.1	16
Courier	24.9	30	3.0	57.9	18.0	2.49	12.4	14
Bulk tanker	86.9	100	200.0	386.9	40.0	0.83	4.1	27
Liner	198.4	60	14.0	272.4	100.0	2.94	14.7	69
Large liner	885.5	300	65.0	1250.5	500.3	3.20	16.0	296

Legs. Earth→Mars, Mars→Earth, Earth→Venus, Earth→Mercury, Earth→Ceres, Mars→Ceres, Earth→Jupiter and Jupiter→Saturn, on the game's Kepler orbits (real eccentricities). The departures are spread evenly over one synodic period: 36 on the first five routes, 18 on the last three, where each departure costs more to compute. That makes 234 departures; with five hulls and four tank sizes, 4680 cases. The phase angle is the destination's lead over the origin at departure. The ship leaves on the origin planet's heliocentric state and meets the destination's: the planet-system spirals are not part of the heliocentric leg, as in the game. Closest solar pass 0.1 AU.

Populations. Each B plan the solver returns is re-propagated at 10^{-12} and kept only if it reproduces (Section 5). The statistics then answer one of two questions, and each table says which:

- **B solved** (how good is a successful refinement): only the cases where B returned a verified plan on A's branch. Table 3 is of this kind.
- **B with fallback** (what a planner that flies A whenever B fails would get): every case where A flies. Where B's solver returns no plan, or one worse than A's, the planner flies A's plan; these are *solver misses* and score zero gain. Tables 4 and 5 lead with this, and give the B-solved mean beside it.

The second never exceeds the first; the gap measures the solver, not the band.

What is compared. For each ship and departure, on a log grid of 44 transfer times (0.05–4 Hohmann times, at least 20 days, at most two years past Hohmann):

1. **Same time:** the propellant of A and B at A's shortest feasible time (B's when only B flies), and the largest distance between the two paths along the way.
2. **Shortest time on a full tank:** the earliest arrival each approach can fly with the standard tank (10% reserve), refined by bisection to 0.02%.
3. **Cheapest trip:** propellant bought at the game's fuel price plus the ship's daily cost times the transfer time. The ship tanks 1.1 times the burn, solved self-consistently since the load changes s_0. The optimum over the grid is refined by a parabola. Done for every tank variant, which gives the best tank per approach.

The *error* of A is its excess over B: propellant at the same time, time on the same tank, cost of the cheapest trip. B is the reference.

5 Results

Same transfer time. Table 3 and Figure 1 compare the propellant at A's shortest full-tank time. That is the point where the band binds hardest on A's plan; at longer times both approaches burn less and the gap narrows (Figure 2). Over 1096 pairs A's excess is between 0.02% and 45%, with a median of 1.0% and a mean of 3.2%.

- **Cargo hulls** (freighter, courier, bulk tanker) lose about 1% on average. Their full-power thrust at 250 km/s (0.8–2.5 mm/s^2) is below what the path asks for most of the way, so A throttles only briefly around the mid-course minimum.
- **Liners** have 3 mm/s^2 at the ceiling. They throttle for much of a transfer, and A loses 6–7% on average and up to 37%.

- **Outer legs** (Earth→Jupiter, Jupiter→Saturn) are the worst, at about 10% on average. Their thrust profile is low for months.

In 72 cases B did not return a verified plan at that time. In 2 cases B's plan burns less than the ideal solver's plan at the same time. Since no banded plan can beat the global ideal optimum, the ideal solver's plan was not that optimum there: B landed on another, better branch. These cases are left out of the statistics.

Table 3: Same transfer time (A's shortest on a full tank). A's excess propellant over B, the share of the transfer B spends coasting, and the largest distance between the two paths. Population: B solved. "no B": no verified B plan. "branch": B landed on a different local optimum, cheaper than the ideal solver's.

Ship	cases	min %	max %	mean %	coast	sep. mean	sep. max	no B	branch
		A's extra propellant over B			of T	(10^6 km)			
Freighter	223	0.04	42.8	0.91	15%	2.4	123.8	11	0
Courier	216	0.08	44.7	1.18	15%	4.9	504.5	16	2
Bulk tanker	221	0.02	9.7	0.95	15%	1.8	57.1	13	0
Liner	220	0.62	36.8	6.91	37%	10.4	307.8	14	0
Large liner	216	0.54	37.4	6.14	35%	9.5	295.0	18	0

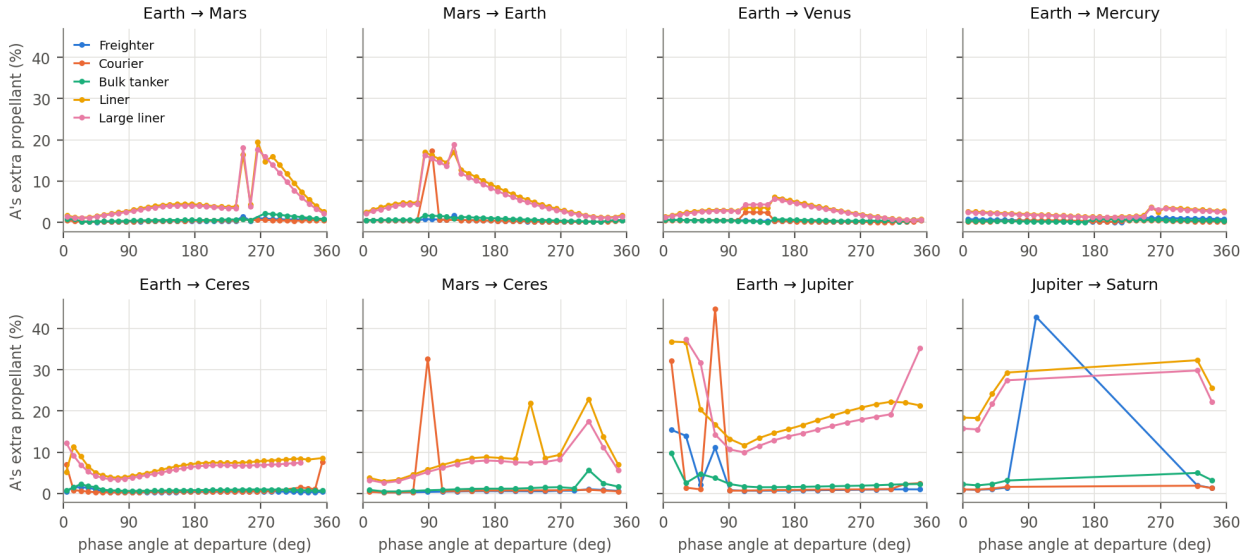


Figure 1: A's extra propellant over B at the same transfer time, against the departure phase angle. The liners' peaks are departures where the shortest time is long and the thrust stays low through the middle of the trip.

Shape of the optimum. B is bang–coast–bang wherever A throttles. Inside the band it follows the ideal thrust law. Where the ideal thrust drops below the ceiling, B burns at full power and 250 km/s, then coasts, then burns again (Figures 5 and 6). Coasting takes about 15% of a cargo transfer and 35–37% of a liner transfer (Figure 3). The floor is almost never touched (below 0.1% of the time in every case): at the times compared here the ideal thrust exceeds the floor only near the ends, and B's continuation turns those into slightly longer ceiling burns.

Trajectory differences. Both plans start and end on the same states, so the difference is in between. On the inner routes the paths separate by a median of 0.4–1.2 million km (up to 46 million km for the liners). On the outer routes the median is 2–9 million km. The largest separations, 100–500 million km, are two-year Jupiter transfers in which B's long coast puts it on a visibly different ellipse (Figure 4). In the game these are not small corrections: a B ship on a map would fly a different line from the one A plans.

Shortest time on a full tank. Table 4 and Figure 7. A's earliest arrival is 4.0% later than B's on average; equivalently, B's trip is 3.4% shorter (3.7% where B solved). The large gains are on the Earth→Venus and Earth→Mercury windows of the cargo hulls: at most A takes 67% longer than B, which is B's trip 40% shorter. There A's shortest time is limited by the 50 km/s floor (the ideal path asks for more thrust than the

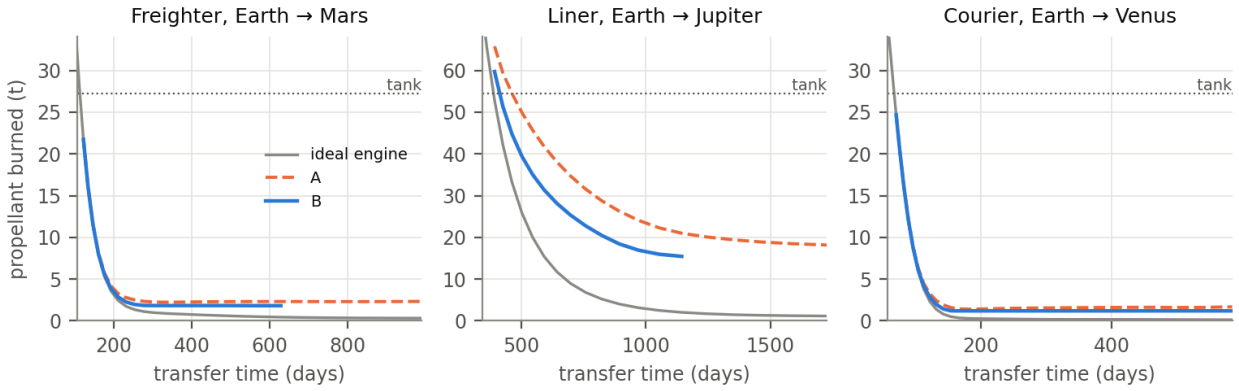


Figure 2: Propellant against transfer time for one departure each: the ideal-engine candidate (the ideal solver's local optimum; a lower bound for A and B on its branch), A and B. Where B's curve ends, its solver stopped converging (long, coast-dominated transfers). The dotted line is the standard tank less the reserve.

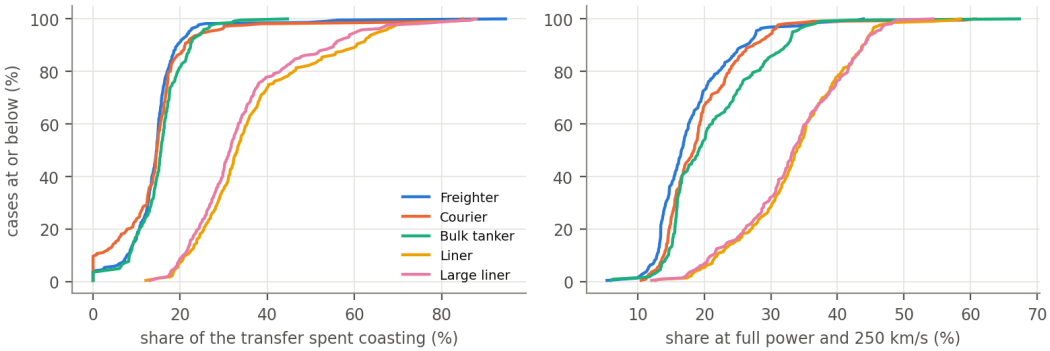


Figure 3: Distribution over departures of the share of the transfer B spends coasting and at full power on the ceiling, at the same time as in Table 3.

drive has near the ends), while B shifts the effort into the middle of the trip. Every departure is flyable by both approaches. B's search fell short of A's time on 88 of the 1170 hull–departure pairs; with fallback those fly A and count as zero.

Table 4: Earliest arrival on the standard tank. Population: B with fallback (a solver miss, where B's search returned a later time than A's or none, flies A and counts as zero); the last column is the B-solved mean. The two middle pairs are the same comparison read both ways: $T_A/T_B - 1$ and $1 - T_B/T_A$.

Ship	departures	solver miss	max % A longer than B	mean % A longer than B	max % B shorter than A	mean % B shorter than A	mean % B solved
Freighter	234	15	67.0	7.34	40.1	5.84	6.24
Courier	234	21	52.3	4.96	34.3	4.05	4.45
Bulk tanker	234	15	48.3	2.21	32.6	1.98	2.11
Liner	234	16	33.9	2.94	25.3	2.68	2.88
Large liner	234	21	58.5	2.62	36.9	2.36	2.59

Cheapest trip and tank choice. With propellant at the game's price and time at each ship's daily cost, the cheapest trip with A costs 1.95% more than with B on average with fallback, 2.19% where B solved (Table 5). That is 1% for cargo and 3% for the liners, up to 31%. At the optimum the two transfer times are within 3% of each other (B is 2–3% shorter for the liners), and A burns 4–11% more. The economics pick long, cheap transfers: cargo ships burn 28–40% of a standard tank and liners about 70%. That is why the half-size tank is the cheapest choice for most cargo departures (Table 6, Figure 8): a smaller tank means less mass to push, and the cargo hulls seldom need more. Banded planning makes small tanks more attractive still. It moves the liners towards the standard and half tanks, because A's throttling penalty grows when a smaller tank makes the ship lighter and raises its thrust at the ceiling. The best tank differs between A and B on 140 of 1170 hull–departure pairs (12%): 22% of the liner departures and 5% of the cargo ones.

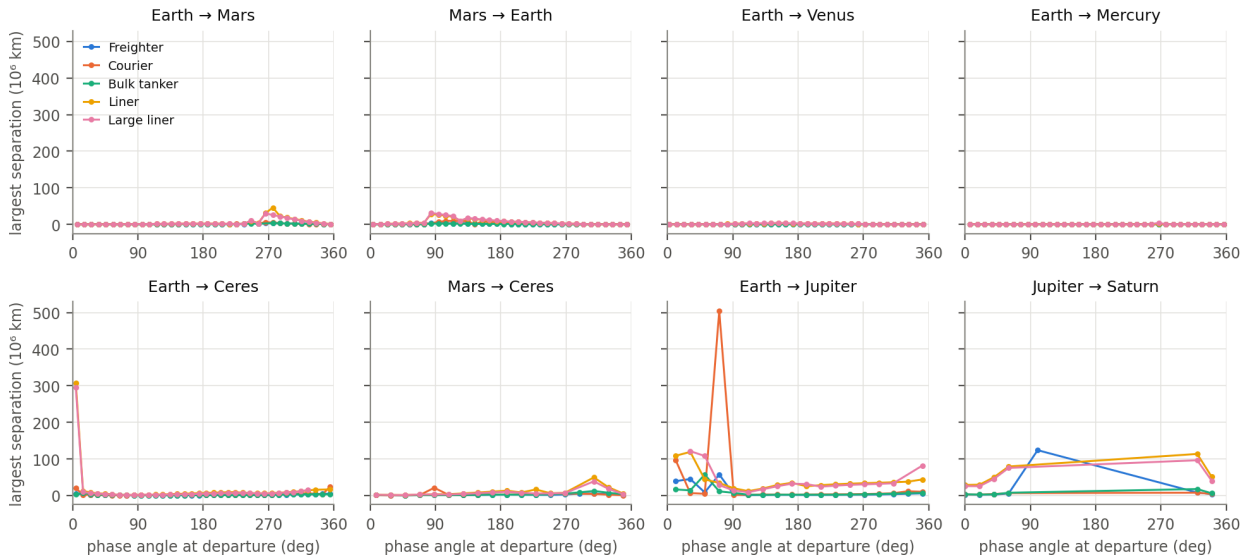


Figure 4: Largest distance between A's and B's paths at the same transfer time.

Table 5: Cheapest trip (fuel bought plus daily cost times transfer time), standard tank. A's extra cost over B, with fallback and where B solved. The T and fuel ratios are averaged over the departures where B's own optimum won.

Ship	cases	solver miss	max % with fallback	mean % with fallback	mean % B solved	T_A/T_B mean, where B's plan won	$\text{fuel}_A/\text{fuel}_B$ mean, where B's plan won
Freighter	234	19	6.25	0.82	0.89	1.009	1.037
Courier	234	21	2.87	1.44	1.58	0.996	1.113
Bulk tanker	234	38	4.61	1.27	1.52	0.996	1.099
Liner	234	24	31.22	3.10	3.46	1.025	1.076
Large liner	234	25	30.43	3.11	3.49	1.027	1.078

Speed. Table 7 and Figure 9, on one thread, for the 240 base-tank departures of a six-departure subset. The task is the planner's: find the earliest arrival for a full tank from scratch. That means the ideal minimum time first, then T up in 5% steps until the approach fits, then bisection.

- **Single solves.** One B solve from the ideal seed takes a median 1.1 ms. A warm solve from a neighbouring time takes 0.5 ms, faster than one ideal solve.
- **Searches.** A full B search takes a median 10 ms, against 3 ms for the ideal engine and 5 ms for A.
- **The tail.** 10% of B searches take more than 114 ms. Searches that fail take 15–74 s each, spent in the smoothing fallback on case after case.

Over the whole study, 16.1% of the 271 903 fixed-time B solves failed. Most of those are at times where the fuel curve has already ended (very long or infeasible), but they are what makes the sweeps slow: the study took 7.9 hours on 13 threads.

Accuracy. Every B plan the solver returns, in the sweeps and the searches alike, is propagated again at a tolerance of 10^{-12}. It is kept only if it arrives within 100 km and 0.1 m/s of the target and its propellant changes by less than 0.5%; otherwise it counts as a failed solve. In the study 1 772 of 228 027 converged plans (0.78%) were rejected; the worst accepted plan misses by 100 km. In the timing set the accepted plans miss by a median of 20 m and at most 44.2 km. Before this check and the interior arc test, one timing plan (a large liner, Earth→Venus) missed by 2.6 million km and 2.9 t of propellant: a short arc grazed its switching boundary, so the arc sequence, and with it the end state, changed under a tighter tolerance. Grazing extremals like this are ill-conditioned for any switching solver; a planner needs the same check.

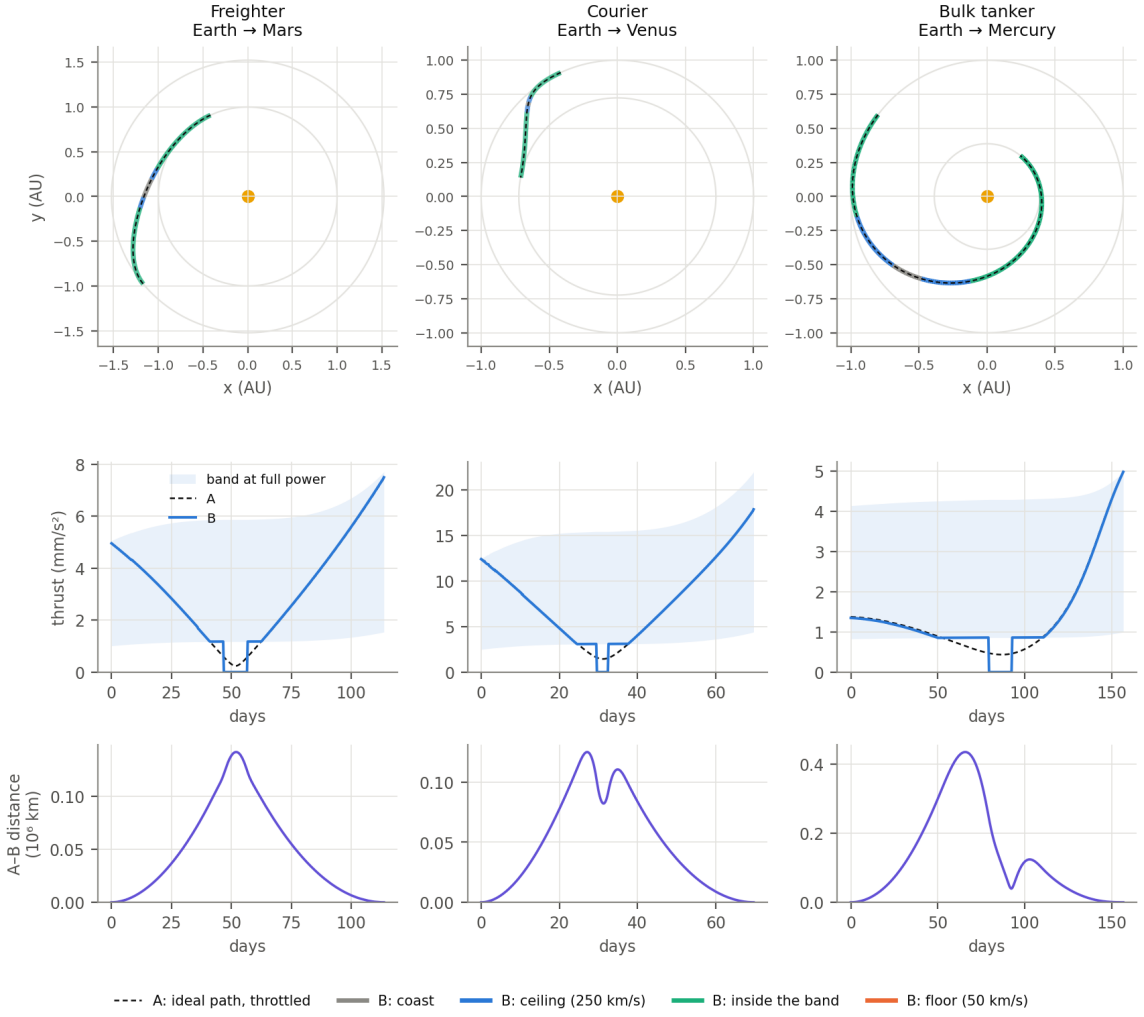


Figure 5: Inner-system examples at A's shortest full-tank time. Top: B's path coloured by arc, with A's dashed. Middle: thrust acceleration against the band at full power (between the 250 and 50 km/s thrusts). Bottom: distance between the two ships. B replaces A's throttled dip with a ceiling burn, a coast and a ceiling burn.

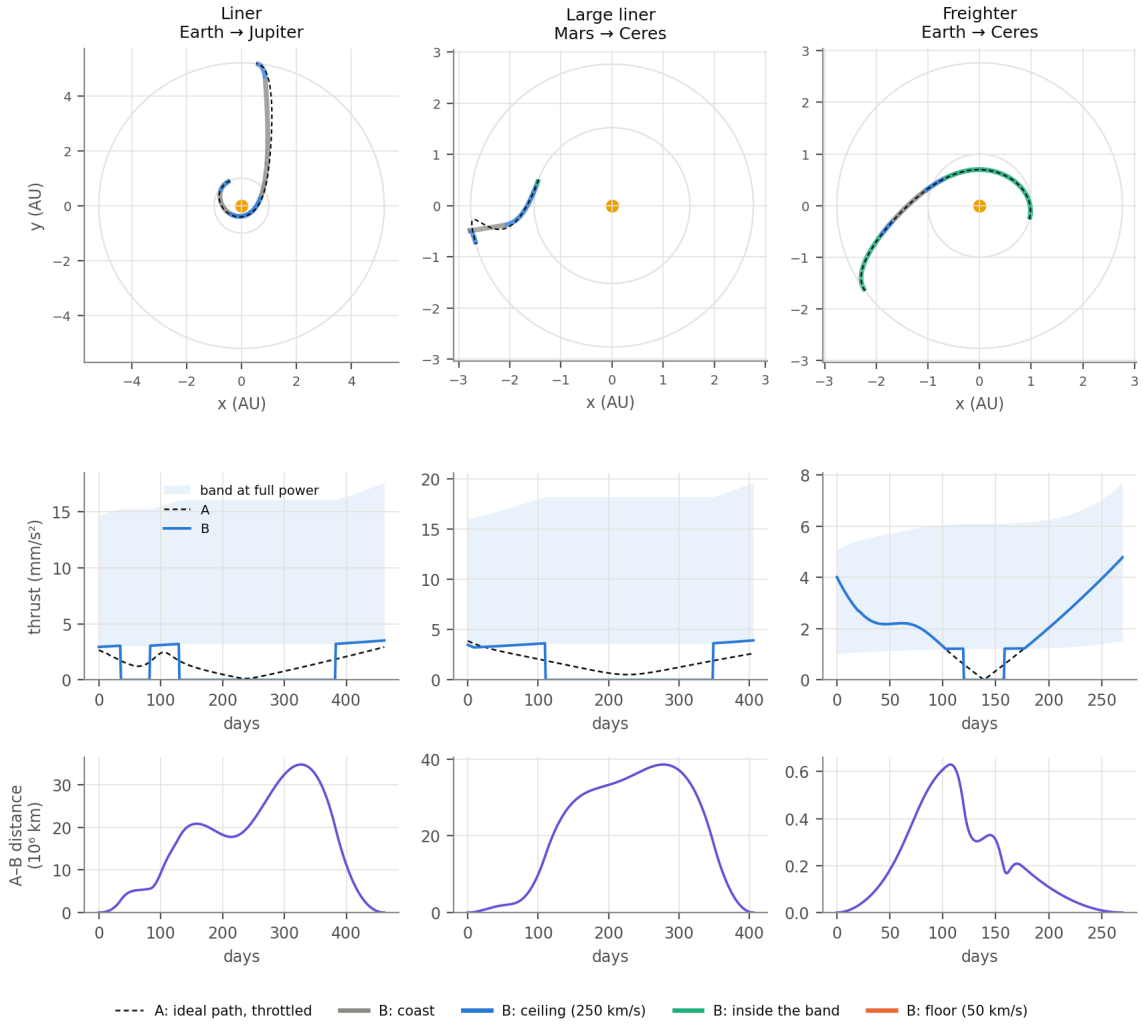


Figure 6: Outer examples. The liner to Jupiter coasts for most of its 460 days. The ships separate by 35–40 million km on the long legs and by under a million km on the freighter's Ceres leg, which stays mostly inside the band. The large liner's Mars→Ceres path turns sharply where its last coast ends: it climbs almost radially and slows to 0.2 km/s near 2.86 AU, so when the final ceiling burn switches on at 2.9 km/s the 3.7 mm/s^2 thrust swings its heading by about 6° a day. The velocity stays continuous (the positions' second differences match gravity plus thrust); A's path, which slows to 2.7 km/s, bends the same way.

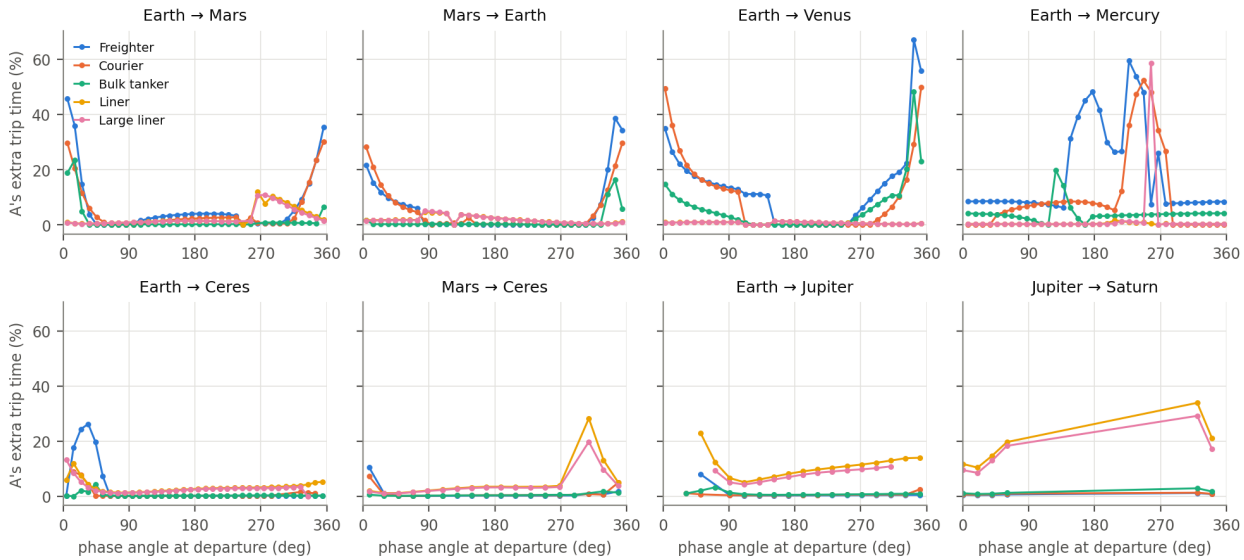


Figure 7: A's extra trip time on a full tank, against the departure phase angle. Solver misses are left out.

Table 6: Number of departures for which each tank size gives the cheapest trip.

Ship	A: best tank				B: best tank				differ
	50%	100%	150%	200%	50%	100%	150%	200%	
Freighter	188	42	2	2	189	42	2	1	11
Courier	219	9	4	2	223	5	3	3	6
Bulk tanker	171	58	3	2	186	42	4	2	20
Liner	39	110	55	30	53	106	56	19	51
Large liner	55	109	62	8	72	119	39	4	52

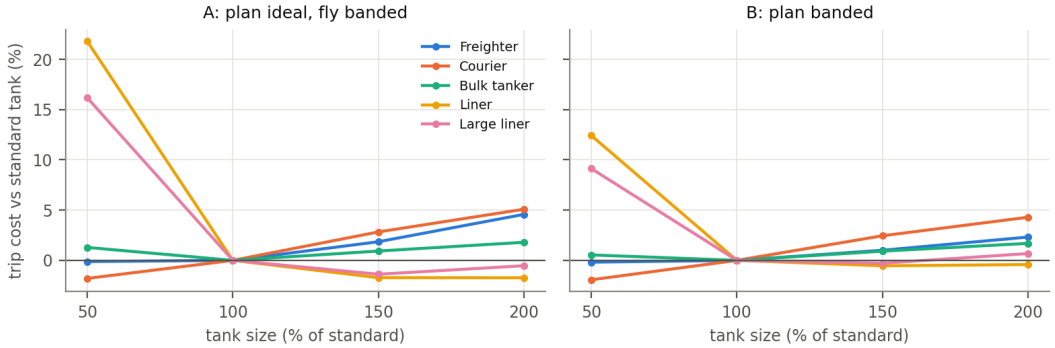


Figure 8: Mean cost of the cheapest trip by tank size, relative to the standard tank. With A, the liners pay heavily for a half tank (20%), because the lighter ship throttles more. With B that penalty halves.

Table 7: Solve times on one thread (ms). "Earliest arrival" is the full search for the shortest time on a full tank. The fixed-T rows time one banded solve at the found time (from the ideal seed) and at 2% longer (warm).

	solved	median	mean	p90	p99	max	over 100 ms
Earliest arrival, ideal engine	240/240	2.5	5.0	11.3	53	72	0
Earliest arrival, A	240/240	5.5	21.2	55.0	248	319	15
Earliest arrival, B	237/240	9.7	163.4	113.6	6061	13079	26
One B solve, from the ideal seed	237/240	1.1	8.3	3.4	177	308	6
One B solve, warm (neighbouring T)	237/240	0.5	1.5	1.0	5	140	1
B search that fails	3		47672			74260	

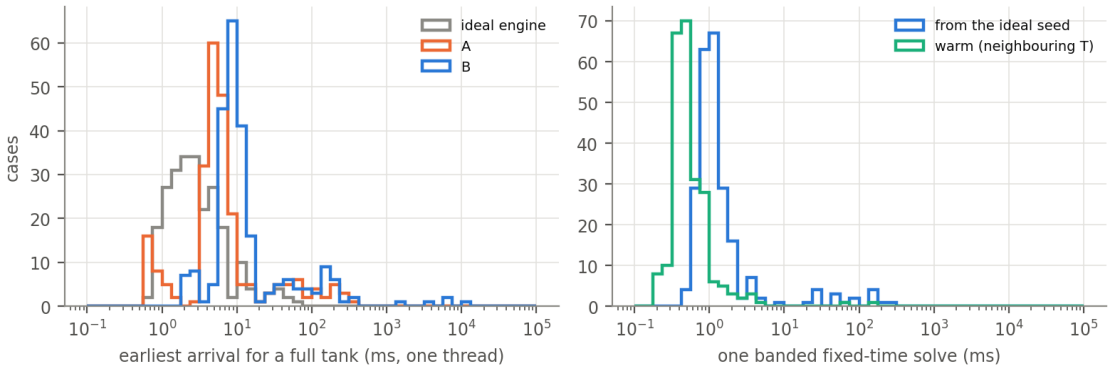


Figure 9: Distribution of the solve times of Table 7.

6 Is B fit for the game?

What B buys. For the cargo hulls, which are most of the fleet's ton-kilometres, about 1% of propellant and 1% of trip cost. That is well inside the noise of the game's economy, where single runs differ by more. For the liners, 6–7% of propellant at their shortest times, 3% of trip cost, and up to 30% on bad departures. For short inner-system windows, a full-tank trip up to 40% shorter (A takes up to 67% longer). B also adds visible behaviour: ships that coast for a third of the trip, and paths that differ from the ideal ones by millions of kilometres.

What B costs.

1. **One atlas no longer serves every ship.** The ideal path does not depend on the ship, so the game stores one atlas of efforts and serves all hulls, loads and porkchops from it. The banded optimum depends on $s_0 = 2P/m_0$, so on the hull, the load and the tank. Porkchops and the planner's caches would need a dimension per ship state, or live solves.

2. **Live solves have a heavy tail.** The median (1 ms per solve, 10 ms per search) is affordable. The failures are not: a failing search with the smoothing fallback can block for a minute. That tail is the study's research configuration, which spends seconds per case on coverage. A game solver would run exact-only and fly A when it fails; the policy benchmark below measures that.

3. **Coverage and conditioning.** 16.1% of fixed-time solves fail across the study, most of the Jupiter→Saturn cases at the compared time, and grazing extremals may not reproduce. Each plan needs a verification propagation and the A fallback behind it.

The refinement policy, benchmarked. The cheap use of B is to refine the trip A picks: same transfer time, same fuel load, one exact-only B solve (no smoothing fallback) from the ideal seed, verified by re-propagation, and A's plan when any step fails. Table 8 runs that policy on every base-tank departure of the study, at A's shortest time and on A's cheapest trip. It solves 94% of the shortest-time cases and 89% of the cheapest trips, and saves 3.0% of propellant and 1.04% of trip cost on average with fallback (up to 2.5%). A solve, verification included, takes a median 2.6 ms; the 99th percentile is 1031 ms and the worst 2997 ms (with the study's threads sharing the machine); 98% of the successful policy solves finish within 50 ms. What the policy cannot do is move the trip: it keeps A's time and tank, so it forgoes the shorter trips and the tank changes of full B. The column "full B" gives the full banded optimum's saving on the same trips for comparison.

Table 8: The refinement policy on the base-tank departures where A flies. Fuel: A's extra propellant over the policy at A's shortest full-tank time. Cost: A's extra trip cost over the policy on A's cheapest trip (same time and load). Means with fallback (a failed policy solve counts as zero). "full B": A's extra cost over the full banded optimum (with fallback), which may change the time. p99: 99th percentile of one policy solve with its check.

Ship	solved	mean %	max %	solved	mean %	full B %	p99 ms
	fuel at A's shortest time			cost of A's cheapest trip			
Freighter	223/234	0.87	42.8	216/234	0.39	0.82	1223
Courier	216/234	1.03	44.7	216/234	1.04	1.44	886
Bulk tanker	221/234	0.90	9.7	201/234	1.05	1.27	939
Liner	219/234	6.46	36.8	206/234	1.27	3.10	802
Large liner	216/234	5.67	37.4	204/234	1.47	3.11	1223

Recommendation (provisional). Keep A as the game's planner, if the band is adopted at all. A is exact (the ship flies what was planned), cheap, keeps the single atlas and the porkchops, and costs the cargo fleet about 1%. Do not make B the planner: the gain does not pay for per-ship atlases and an unreliable tail.

If coasting ships or the liners' 3% matter for the game, the defensible use of B is an *optional refinement of a chosen trip*:

1. The dispatcher picks the trip with A, as now.

2. The departing ship then runs one warm, exact-only B solve at that transfer time (about 1 ms, capped at about 50 ms), seeded from the ideal solution.

3. The ship verifies the plan by re-propagation and flies it, or flies A if any step fails.

The ship then burns no more than planned, often less, and liners would visibly coast. The cost is self-consistency: porkchops and dispatch would show A's propellant, a slightly pessimistic figure, and the trip choice would ignore B's saving. That is not negligible for every ship: the cheapest tank differs on 22% of the liner departures, and B's shortest trips are much shorter on some inner windows. A small average change in T does not mean the choices rarely change. A fully self-consistent banded game, with porkchops in B's numbers, would need per-hull banded atlases. On this evidence they are not worth building for the cargo fleet; for the liners it is a closer call, and depends on how much the game values their trip times and tank choices.

The band itself. The study also says where the band matters: wherever the thrust at 250 km/s is high compared with what the path needs. That is high power for the mass: the liners, and any ship made lighter, such as one with a smaller tank or running empty. A better engine tier with a higher ceiling lowers that thrust and shrinks A's penalty. A tier with a lower floor helps only the shortest, floor-limited inner-system trips.

7 Limitations

- Planar, Sun only, as the game. Planet-system spirals are outside the heliocentric legs; a banded spiral is the throttled one in closed form (the spiral's thrust is constant).
- The ship starts and ends on the planets' states. The game's stations and launch windows add their own constraints.
- The cheapest trip uses the ship's daily cost as the time value. The game's dispatch also weighs cargo value and passengers' time, which push towards shorter trips, where the band matters more.
- B's coverage is not complete. Where neither the exact nor the smoothed continuation converges (long, coast-dominated outer transfers), B flies A's plan, which understates B's advantage there.
- The same-time comparison is made at A's shortest full-tank time, the hardest point for A. At longer times both approaches burn less and the gap narrows (Figure 2).
- Each optimum is local. Both solvers start from the ideal path's branch; other revolution counts and branches were not searched.